Southern Cross University

Final part/chapter of the Autonomic Nervous System Pharmacology

Introductory Pharmacology
PHAR2001

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Objectives

- Explain the difference between direct-acting and indirect-acting Parasympathomimetic drugs
- Know terms used to describe the parasympathomimetic drugs, for instance, muscarinic drugs, cholinergic agents, acetylcholine agonists and antagonists.
- Describe the actions interactions and side effects of the:
  - Direct acting cholinergic agonists bethanechol and pilocarpine
  - Indirect acting cholinergic agonists neostigmine and donepezil
  - The antimuscarinic agents, using atropine as an example.

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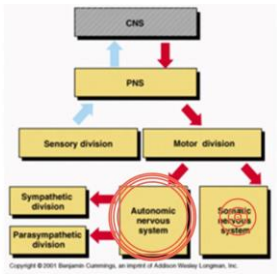
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Drugs of the Parasympathetic Nervous System (PNS)

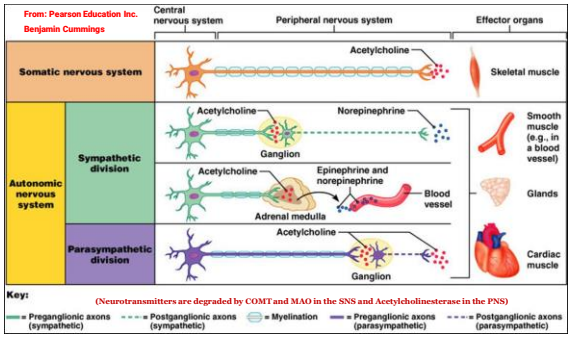
Lesson 1:

Understanding Acetylcholine, the neurotransmitter for the PNS (and the muscles – how can that be?)

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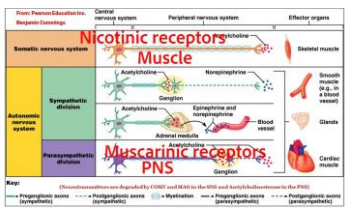
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Acetylcholine binding

Skeletal muscle = nicotinic receptors.
PNS= muscarinic receptors



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The parasympathetic nervous system

* Uses acetylcholine as the neurotransmitter

* Uses primarily muscarinic receptors

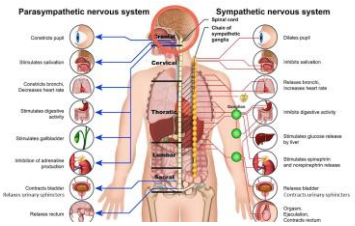
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Different names for the parasympatholytic agents

* Antimuscarinic agents
* Anticholinergic agents
* Parasympathetic antagonists
* Acetylcholine antagonists

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Anticholinergic agents that break into the brain are associated with dementia



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Drugs of the Parasympathetic Nervous System (PNS)

Lesson 2:

Important foundational information about the PNS

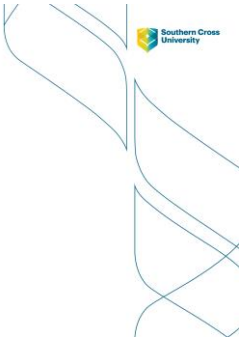
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The parasympathetic nervous system

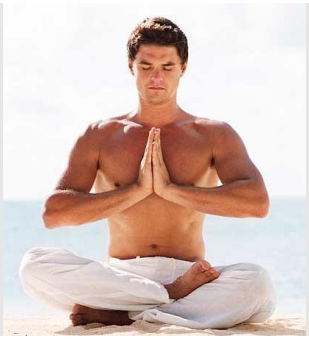
The *'rest and digest'* system

- Heart rate slows down
- Digestion begins
- Pupils constrict

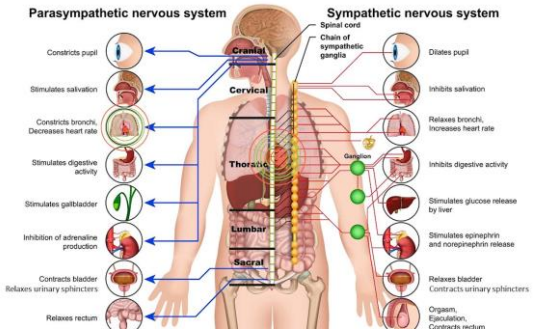
Think about the last time you were able to relax



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Muscarine is a non-selective agonist of the muscarinic acetylcholine receptors

Antidote: a muscarinic antagonist (for instance, atropine)

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Cholinergic agents

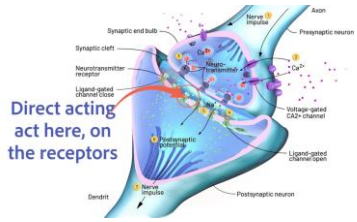
Direct acting or Indirect acting

Can be:

- **Nicotinic**
- **Muscarinic (5 subsets)**
  - **M1 brain (decreased in Alzheimer's disease)**
  - **M2 heart (cause decreased rate and force of heart beat)**
  - **M3 relaxation of GIT, lacrimal secretion, pupil constriction, increased micturition and defecation, bronchoconstriction**
  - **M4 and M5 (not elucidated)**

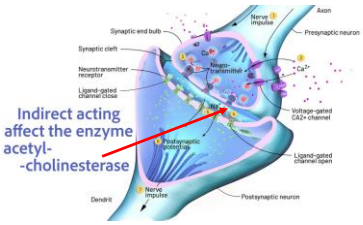
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Direct acting cholinergic agents act on the receptors, acting like acetylcholine

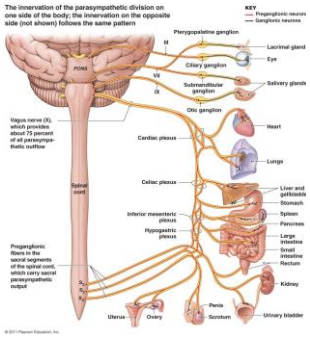


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Indirect acting cholinergic agents act on the enzyme that degrades acetylcholine



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Thanks to Pearson Education 2011

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Cholinergic agents

Direct acting or Indirect acting

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  - **M4 and M5 (not elucidated)**

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Drugs of the PNS
Lesson 3:
Understanding the difference between direct and indirect acting cholinergic agents

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Cholinergic agonists

Direct acting non-selective agonists include:

- **Pilocarpine** (know this drug; used to treat glaucoma. While this is a non-selective agonist, pilocarpine is administered topically and has little systemic absorption, therefore the actions are fairly limited to the m3 receptors in the eyes)
- **Bethanechol** (know this drug; previously used for non-obstructive urinary retention. Adverse effects: bradycardia, hypotension, diarrhoea, all consistent with parasympathetic stimulation.

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Direct acting cholinergic agonists

Indications include non-obstructive postoperative and postpartum urinary retention

Topically in eye to reduce intraocular pressure of glaucoma, mydriasis

Adverse effects

- Extreme caution, even if used in eye
- Bradycardia,
- Hypotension,
- Nausea, vomiting,
- Diarrhoea, dehydration
- Bronchoconstriction



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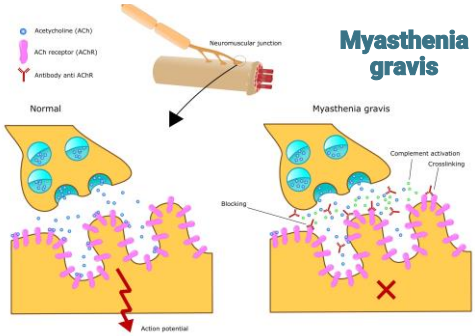
Drugs of the PNS
Lesson 4:
Indirect acting acetylcholine inhibitors for Myasthenia gravis and Alzheimer's disease

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Indirect acting cholinergic agonists

- Act on acetylcholinesterase at the post synaptic cleft
  - Prevents the breakdown of acetylcholine
    - More acetylcholine remains longer
  - Drugs cause a reversible bond; **nerve gas is irreversible**
- Used for patients with:
  - Myasthenia gravis
    - (Edrophonium is used to diagnose myasthenia gravis)
    - **Neostigmine** (know this drug) and pyrido**stigmine** are used to treat MG
  - Alzheimer's disease
    - **Donepezil** (know this drug)
    - **Galantamine** (know this drug)
    - Riva**stigmine**

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Myasthenia gravis

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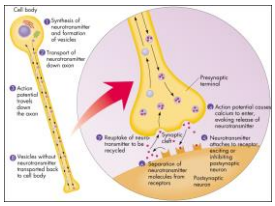
Myasthenia Gravis often affects facial muscles



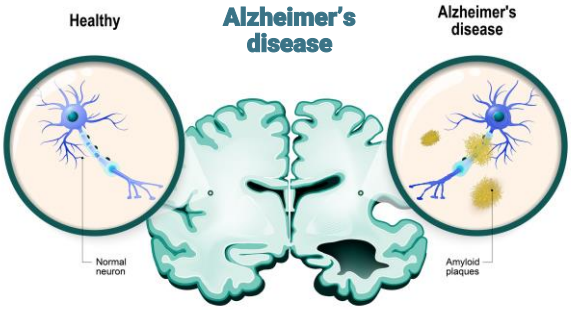
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Myasthenia Gravis

- Myasthenia gravis
  - Autoimmune disorder
  - Destruction of ACh receptors
- Neostigmine
  - Prevents breakdown of ACh
- Side effects:
  - Bradycardia
    - Normally taken with a parasympatholytic such as atropine.
  - Anorexia, nausea, vomiting, abdominal cramps, and diarrhoea



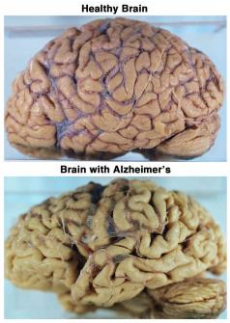
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Alzheimer's disease

- Progressive loss of ACh neurons
- Galantamine
  - Acetylcholinesterase inhibitor
- Git complaints



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Advanced Alzheimer's disease

- **Donepezil** (Aricept; know this drug)
  - Reversible acetylcholinesterase inhibitor that is 100% bioavailable and crosses the BB barrier
  - Two clinical trials indicate it may be effective for moderate to severe Alzheimer's disease
- Should be used with caution in patients with cardiac disease, cardiac conduction disturbances, chronic obstructive pulmonary disease, asthma
- Patients with gastrointestinal disorders should use caution because nausea or vomiting may occur.

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Drugs of the PNS
Lesson 5:
Anticholinergic agents

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Anticholinergic agents

Other terms used:

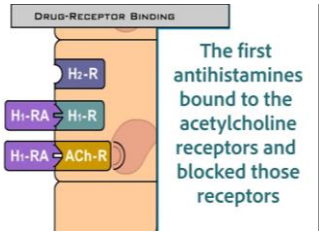
- Anticholinergics
- Parasympatholytics
- Muscarinic antagonists
- Antimuscarinic agents

Allows the sympathetic system to dominate



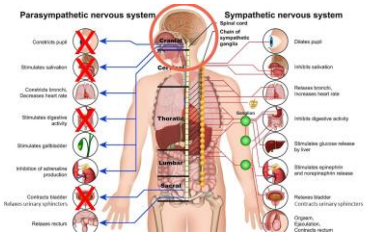
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Many of our medications inadvertently bind to and block the acetylcholine muscarinic receptor



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Anticholinergic agents allow the sympathetic nervous system to dominate



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Muscarinic antagonists

Include
- **Atropine (know this drug)**
- Hyoscine
- Hyoscamine
- Cyclopentolate

Indications:
- Antispasmodics in GIT spasm,
- Counterbalance bradycardia
- Antiemetic
- Antiasthma agent
- Treatment of urinary frequency



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Deadly nightshade contains atropine that is an anticholinergic agent



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This is a common mnemonic for anticholinergic overdose

To describe the physiologic manifestations of atropine overdose:

"Hot as a hare, blind as a bat, dry as a bone, red as a beet, and mad as a hatter" describing

- warm, dry skin from decreased sweating,
- blurry vision, decreased lacrimation,
- vasodilation, and
- central nervous system effects on muscarinic receptors

This set of symptoms is known as anticholinergic toxidrome, and may also be caused by other drugs with anticholinergic effects, such as hyoscine hydrobromide (scopolamine), diphenhydramine, phenothiazine antipsychotics and benztropine

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References

- Goldsy RA et Al. Immunology 5th ed. WH Freeman Publishers, 2003
- Marieb E. Human Anatomy and Physiology, 5th ed. Benjamin Cummings, San Francisco, 2001
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